Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering
09/2023 – Expected 06/2028

- GPA: 2.93/4.00
- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi; Regional Organizational Team Member at ESTIEM; Work and Travel USA participant; Student-Athlete (sailing, skiing & windsurfing), doing wakeboarding, kiteboarding, surfing, bouldering, running, fitness, and traveling as hobbies.
- Volunteer: Volunteering Division Member at Industrial Engineering Club (BÜYAK); provided educational mentorship to high school students.

Kabataş Boys' High School
09/2018 – 06/2023

- GPA: 3.78/4.00
- Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club.
- Volunteer: Community Involvement Projects Club and TOÇEV (Tüvana Okuma İstekli Çocuk Eğitim Vakfı).

Experience

Backend Developer Intern at DefineX
09/2026 – Present

- Joining the backend development team to contribute to scalable enterprise software solutions, robust backend architecture, RESTful API design, and system optimizations in a dynamic enterprise environment.

Site Reliability Engineering Intern at Akbank
07/2026 – 09/2026

- Completed a 2-month internship with the SRE team, contributing to the reliability, scalability, and performance of mission-critical banking systems. Engaged in Linux systems engineering, operational monitoring, and high-availability systems management.

Software Engineering Intern at Smash Mate
06/2026 – 07/2026

- Completed an intensive 1-month internship during the critical pre-launch phase, contributing to a complex microservices-oriented backend (Java, Quarkus) and a mobile frontend application (React Native). Developed core backend features, resolved high-priority issues, and authored comprehensive unit/integration tests to ensure zero-defect deployments. Played an active role in agile workflows, code reviews, and system optimizations, leading directly to the product's successful official launch.

Technical Skills & Projects

- **AlphaTrigger** (Python, FastAPI, Next.js, WebSockets) — Autonomous trading simulator with Z-Score mean reversion & volatility sizing.
- **FinTech Platform** (Python, FastAPI, Next.js) — Real-time dashboard with limit-order matching engine & portfolio risk analytics.
- **Seated** (Next.js 15+, TypeScript, Prisma, PostgreSQL) — Multi-tenant SaaS for real-time restaurant floor plans, reservations & operations.
- **Performo** (TypeScript, Next.js, Express, Tailwind CSS) — Administrative console with role-based access control (RBAC) & task workflows.
- **Enterprise FAQ** (Next.js, TypeScript, Express, SWR) — Enterprise knowledge base with optimistic UI updates, caching & error boundaries.
- **Space Shooter IPC** (C11, POSIX FIFOs, Signals, dlopen) — Asynchronous 60 FPS input engine using non-blocking FIFOs and signals.
- **SchedSim** (x86-64 GNU Assembly, Linux Syscalls) — Low-level CPU scheduling simulator executing FCFS, SJF, SRTF, and Round Robin.
- **Solar Escape** (RISC-V RV32I Assembly, Spike) — Bare-metal pathfinding algorithm with greedy route selection & manual stack framing.
- **CoreMark Profiling** (RISC-V, CoreMark) — Benchmarking study assessing compiler optimizations (-O0 to -Ofast) and ISA pipeline throughput.
- **Spireslayer** (C, POSIX, Makefile) — Unix command interpreter featuring a custom backtracking recursive descent parser.
- **Aura** (Python) — Functional programming language interpreter supporting closures, lexical scoping, and currying.
- **MatrixNet** (Java, Graph Theory) — Network topology analysis tool implementing Tarjan's SCC algorithm & articulation points.
- **GigMatch-Pro** (Java, Priority Queues) — Algorithmic job-matching engine leveraging max-heaps for priority-based candidate allocation.
- **Smash Minesweeper** (React Native, TypeScript, Expo) — Mobile puzzle game built with reducer-driven state engine and BFS flood-fill.
- **Qt 2048** (C++, Qt6, WebAssembly) — Cross-platform game compiled to WebAssembly for browser deployment with state undo.
- **Nightpass** (Java) — Terminal-based strategy survival card game driven by a deterministic finite state machine.

Tools & Languages

- **Tools & Software:** Git/GitHub, AI-Assisted Development, MS Office
- **Languages:** Turkish (Native), English (Advanced), German (Beginner)